1. **Download and install Tableau Public or Microsoft Power BI Desktop.**

```
import pandas as pd
import matplotlib.pyplot as plt
```

2. **Import a real-world dataset such as the Superstore Sales Dataset, Global Superstore Dataset, or Amazon Sales Dataset.**

```
import pandas as pd
data = {
    "Order ID": [
        "ORD001", "ORD002", "ORD003", "ORD004",
        "ORD005", "ORD006", "ORD007", "ORD008"
    ],
    "Customer ID": [
        "CUS001", "CUS002", "CUS003", "CUS004",
        "CUS005", "CUS006", "CUS007", "CUS008"
    ],
    "Order Date": [
        "2025-01-05", "2025-01-12", "2025-02-03", "2025-02-15",
        "2025-03-10", "2025-03-22", "2025-04-08", "2025-04-20"
    ],
    "Region": [
        "North", "South", "East", "West",
        "North", "South", "East", "West"
    ],
    "Category": [
        "Technology", "Furniture", "Technology", "Office Supplies",
        "Furniture", "Technology", "Office Supplies", "Technology"
    ],
    "Sales": [
        12000, 8500, 15000, 7000,
        9500, 11000, 6500, 13500
    ],
    "Profit": [
        2500, 1200, 3200, 900,
        1400, 2200, 800, 2800
    ],

    "Orders": [
        80, 55, 95, 48,
        60, 72, 42, 88
    ]
}
df = pd.DataFrame(data)
df["Order Date"] = pd.to_datetime(df["Order Date"])
df
```

	Order ID	Customer ID	Order Date	Region	Category	Sales	Profit	Orders
0	ORD001	CUS001	2025-01-05	North	Technology	12000	2500	80
1	ORD002	CUS002	2025-01-12	South	Furniture	8500	1200	55
2	ORD003	CUS003	2025-02-03	East	Technology	15000	3200	95
3	ORD004	CUS004	2025-02-15	West	Office Supplies	7000	900	48
4	ORD005	CUS005	2025-03-10	North	Furniture	9500	1400	60
5	ORD006	CUS006	2025-03-22	South	Technology	11000	2200	72
6	ORD007	CUS007	2025-04-08	East	Office Supplies	6500	800	42
7	ORD008	CUS008	2025-04-20	West	Technology	13500	2800	88

Next steps: Generate code with df | New interactive sheet

3. **Verify the imported data by checking field names, data types, and relationships between tables (if applicable)**.

```
print("Dataset Shape:", df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Dataset Shape: (8, 8)

Column Names:
Index(['Order ID', 'Customer ID', 'Order Date', 'Region', 'Category', 'Sales',
       'Profit', 'Orders'],
      dtype='object')

Data Types:
Order ID               object
Customer ID            object
Order Date     datetime64[ns]
Region                 object
Category               object
Sales                   int64
Profit                  int64
Orders                  int64
dtype: object

Missing Values:
Order ID       0
Customer ID    0
Order Date     0
Region         0
Category       0
Sales          0
Profit         0
Orders         0
dtype: int64
```

4. **Create Key Performance Indicators (KPIs) such as Total Sales, Total Profit, Number of Orders, Average Order Value, orCustomer Count.**

```
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_orders = df["Order ID"].nunique()
customer_count = df["Customer ID"].nunique()

average_order_value = total_sales / total_orders

print("Total Sales:", total_sales)
print("Total Profit:", total_profit)
print("Number of Orders:", total_orders)
print("Customer Count:", customer_count)
print("Average Order Value:", average_order_value)
```

```
Total Sales: 83000
Total Profit: 15000
Number of Orders: 8
Customer Count: 8
Average Order Value: 10375.0
```

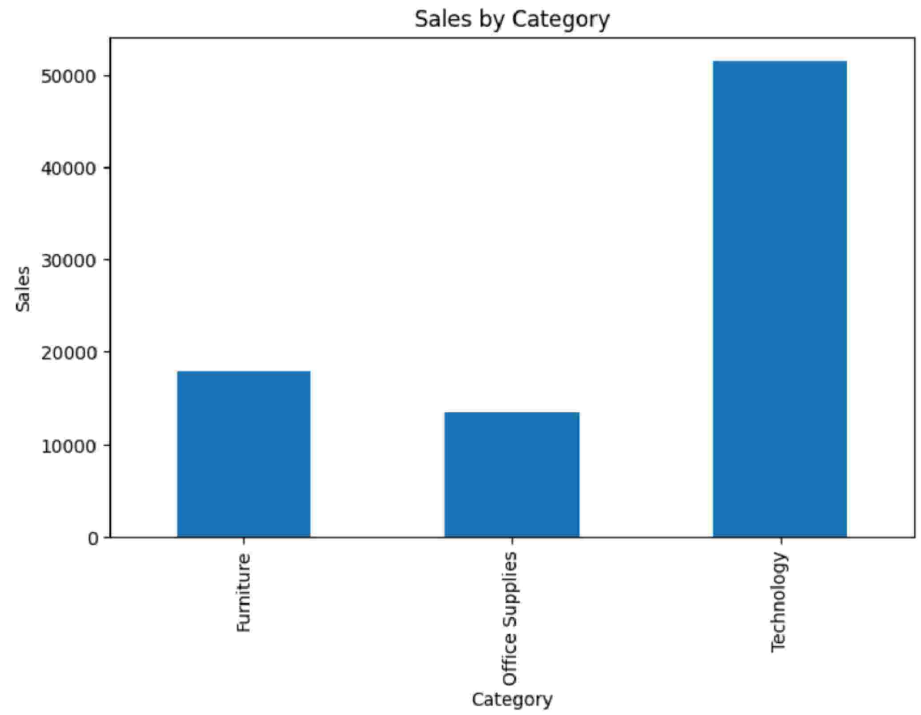
5. **Develop multiple visualizations, such as bar charts, line charts, pie charts, maps, and treemaps, to analyze businessperformance from different perspectives.**

Sales by Category

```
category_sales = df.groupby("Category")["Sales"].sum()
```

```
category_sales.plot(kind="bar", figsize=(8,5))

plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()
```

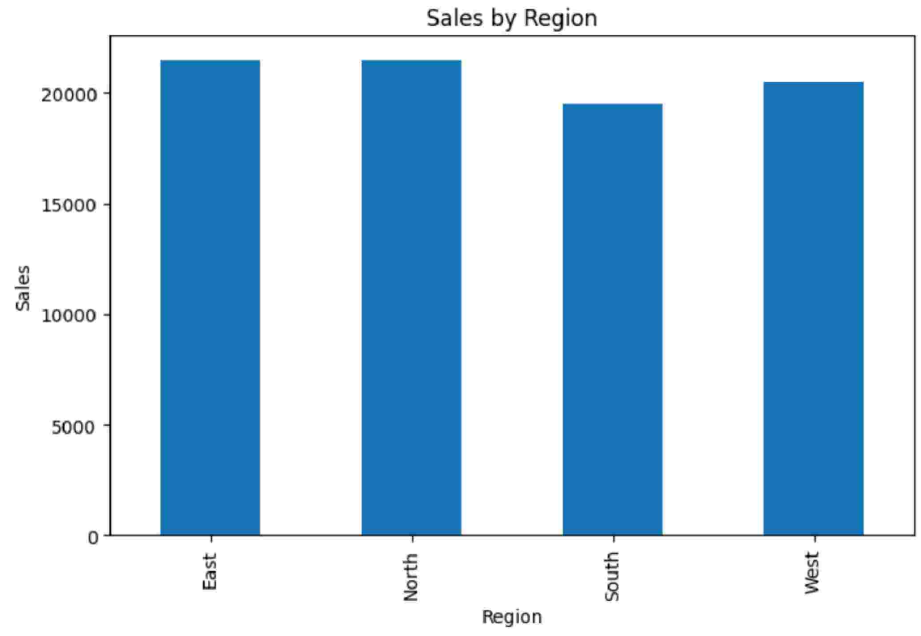


Sales by Region

```
region_sales = df.groupby("Region")["Sales"].sum()

region_sales.plot(kind="bar", figsize=(8,5))

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")
plt.show()
```



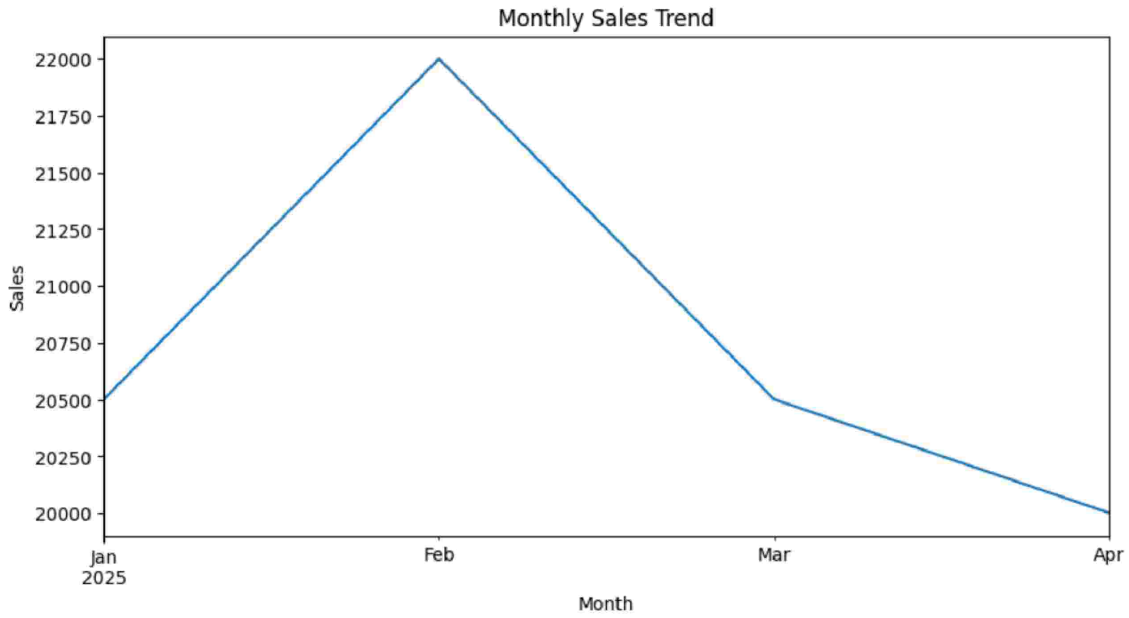
Sales Trend

```
df["Order Date"] = pd.to_datetime(df["Order Date"])

monthly_sales = df.groupby(
    df["Order Date"].dt.to_period("M")
)["Sales"].sum()

monthly_sales.plot(figsize=(10,5))

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```



6. **Add interactive filters, slicers, and drill-down features to allow users to explore the data by region, product category,customer segment, or time period.**

```
technology = df[df["Category"] == "Technology"]

display(technology.head())
```

	Order ID	Customer ID	Order Date	Region	Category	Sales	Profit	Orders
0	ORD001	CUS001	2025-01-05	North	Technology	12000	2500	80
2	ORD003	CUS003	2025-02-03	East	Technology	15000	3200	95
5	ORD006	CUS006	2025-03-22	South	Technology	11000	2200	72
7	ORD008	CUS008	2025-04-20	West	Technology	13500	2800	88

Filter by region:

```
west_region = df[df["Region"] == "West"]

display(west_region.head())
```

	Order ID	Customer ID	Order Date	Region	Category	Sales	Profit	Orders
3	ORD004	CUS004	2025-02-15	West	Office Supplies	7000	900	48
7	ORD008	CUS008	2025-04-20	West	Technology	13500	2800	88

7. **Arrange all visualizations into a professional dashboard with appropriate titles, formatting, and layout.**

```
print("========== SALES DASHBOARD ==========")
print("Total Sales:", round(total_sales, 2))
print("Total Profit:", round(total_profit, 2))
print("Total Orders:", total_orders)
print("Customer Count:", customer_count)
print("Average Order Value:", round(average_order_value, 2))
```

```
========== SALES DASHBOARD ==========
Total Sales: 83000
Total Profit: 15000
Total Orders: 8
Customer Count: 8
Average Order Value: 10375.0
```

8. Analyze the dashboard to identify business trends, top-performing products, high revenue regions, and areas requiringimprovement.

```
top_category = category_sales.idxmax()
top_region = region_sales.idxmax()

print("Top Performing Category:", top_category)
print("Top Performing Region:", top_region)
```

```
Top Performing Category: Technology
Top Performing Region: East
```

9. Publish or export the dashboard for presentation and sharing with stakeholders.

```
print("========== KEY BUSINESS INSIGHTS ==========")

print("1. The highest-selling category is:", top_category)
print("2. The highest-revenue region is:", top_region)
print("3. Total sales are:", round(total_sales, 2))
print("4. Total profit is:", round(total_profit, 2))
print("5. Total number of orders is:", total_orders)
```

```
========== KEY BUSINESS INSIGHTS ==========
1. The highest-selling category is: Technology
2. The highest-revenue region is: East
3. Total sales are: 83000
4. Total profit is: 15000
5. Total number of orders is: 8
```

10. RESULT:

An interactive business dashboard was developed using the sales dataset. KPIs, charts, filters and business insights were generated to analyze sales performance and support data-driven decision-making.